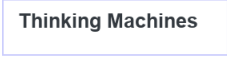


Experience Bench V1 and V2

An evaluation and data framework for multimodal assistants and agents. It measures whether systems can carry visual evidence forward, retain user instructions, maintain a coherent world state and act appropriately across natural multi-turn video interactions.

3 interaction axes	3-8 interaction turns	Video evidence required
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Target customers

Model + device manufacturers Own the assistant, model and visual hardware surface.	     
Frontier multimodal labs Need post-training data and reproducible evaluations for visual interaction.	

\$58M TAM (2027)

Video Data for Interaction Models for Wearables

\$58M

annual external data + evaluation spend

\$6.4B displayless smart-glasses TAM (IDC - smart-glasses forecast) × R&D share × **15%** data/evals × vendor share

Scenario	R&D	External	TAM
Low	10%	30%	\$29M
Base	15%	40%	\$58M
High	20%	50%	\$96M

Data/evaluation and vendor shares are planning assumptions.

Experience Bench V1

The framework measures memory, instruction retention and self-coherence across interactions where visual evidence changes over time.



AXIS 01

Inference Memory

Infer a fact from earlier visual and multimodal evidence, retain it, and apply it later.

Car inspection

A person points to distorted parts, damaged wiring and leaks while giving underspecified comments such as “this is bad.”

“List every problem we found, where it was located, and which ones I considered urgent.”

- correct referent
- retains inferred severity
- complete defect list
- no transcript-only shortcut

AXIS 02

Instruction Retention

Continue applying an earlier visual-monitoring instruction and all of its constraints throughout the interaction.

Restaurant site survey

A restaurant owner is scouting a location for a new restaurant, while walking around the neighborhood inspecting crowds at current restaurants

“Log every sit-down restaurant as empty, quiet, busy or full. Exclude cafés and takeout counters. Don’t interrupt our conversation, just make note of it.”

The user changes topics, revisits streets and passes ambiguous establishments without repeating the instruction.

- eligible venues covered
- exclusions obeyed
- capacity rubric retained
- correct response timing

AXIS 03

Self-Coherence

Maintain a consistent, evolving representation of the scene and revise it only when new visual evidence warrants a change.

Trip packing

The user gives an itinerary, then moves, removes and replaces clothing across multiple suitcases.

“Before I leave, tell me what is missing, what was replaced, and which suitcase contains each required item.”

- correct state transitions
- no contradictions
- replacement recognized
- final location grounded

Experience Bench V2

Extends Experience Bench V1 into an agentic evaluation with a simulated mobile or wearable environment and app tools. The agent must use visual evidence from the world to decide whether, when and how to take actions while preserving user intent and avoiding unintended side effects.

Inputs Continuous video, ambient audio, dialogue, sensor data and current app state.
Available actions Wait, alert, ask, save, edit, remove, organize and reconcile records.
Evaluation Reasoning accuracy, action timing, final app state and unintended side effects.

INFERENCE MEMORY

Apartment search

The user tours several apartments and reacts naturally to sunlight, noise, storage and layout without stating a complete preference list.

“Save the best two and make me a comparison I can review tonight.”

Actions: save listings in Maps; create a comparison in Notes; attach the observations supporting each choice.

INSTRUCTION RETENTION

Running assistant

The user defines a monitoring rule before starting, then encounters hills, brief stops, sensor spikes and street crossings.

“If my heart rate stays above Zone 4 while my pace declines for two intervals, tell me to slow down and log where it happened.”

Actions: monitor Health and Workout; issue one timely alert; mark the event and location in the workout record.

SELF-COHERENCE

Pantry and budget management

The model observes the pantry before shopping, then tracks items that are added, removed, substituted or discovered to already be at home.

“Get everything we need for the week, but keep the basket under \$60.”

Actions: reconcile Pantry and Shopping List; track the live cart total; remove duplicates; propose substitutions before exceeding the budget.

COLLECTION PIPELINE

Collection methodology

0. Proof-of-concept Start with existing online videos to bootstrap initial interaction samples.	1. Blueprint Propose axis-specific strategies that expose visual interaction failures.
2. Record Participants record videos with rich scenes, evolving context and natural dialogue.	3. Verify Create evaluation cases and rubrics that produce reproducible interaction findings.